

PoreAccess-CO₂: From Random Packed-Bed Structure to a Fast Adsorption Model

Research concept proposal
Ali Khalifa and Heiko Briesen
Technical University of Munich, Chair of Process Systems Engineering

Abstract

Packed beds are used for CO₂ adsorption, gas purification and drying. Even when two beds contain the same monodisperse beads and have nearly the same porosity, their void pathways can differ. These differences may control how easily gas enters the bed and reaches the adsorbent, but they are hidden in conventional one-dimensional models. This project will connect three model levels. DEM will generate mechanically realistic zeolite beds. A pore-network model will resolve diffusion and adsorption in each void structure. A simple homogeneous model will then reproduce each uptake curve using one fitted effective diffusivity. The central test is whether a physically defined inlet-accessibility descriptor can predict this diffusivity for a new packing. The expected outcome is an interpretable structure-to-model closure: particle geometry in, fast transient adsorption prediction out. The study deliberately isolates diffusion-driven adsorption; no CFD or imposed gas flow is required in the first step.

1 Motivation and research question

Fast bed-scale models normally describe the packing by bulk porosity and an effective transport coefficient. Porosity measures how much void space exists, but not whether this space forms open paths from the inlet into the bed. A local bottleneck near the inlet may therefore slow adsorption even when the overall porosity is unchanged. Resolving every pore can reveal this effect, but it is too detailed for repeated process calculations.

The central question is:

Can a simple pore-accessibility measure translate a random particle packing into the effective diffusivity required by a fast one-dimensional CO₂ adsorption model?

Figure 1 shows the proposed physical problem. CO₂ enters a settled zeolite bed from a reservoir below. It diffuses through connected voids and is adsorbed by the beads. Only the random packing realization changes.

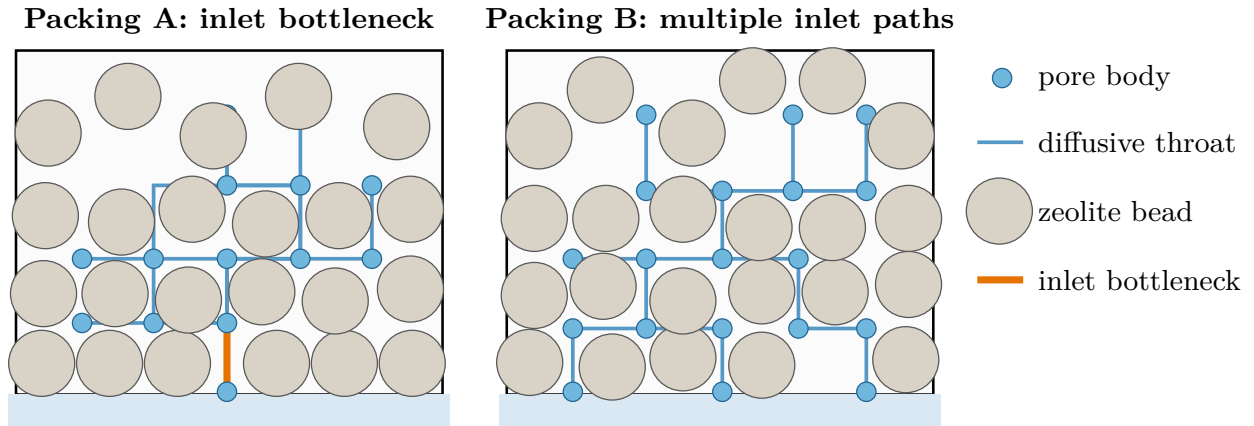


Figure 1: Two illustrative monodisperse beds with different random arrangements. Blue lines indicate the connected pore network. In Packing A, the closely packed bottom layer leaves one narrow entrance (orange), so all transport must first pass through this bottleneck. Packing B has several independent inlet connections and parallel routes. The networks are schematic working hypotheses to be tested.

2 Hypothesis and objectives

The working hypothesis is:

Beds with similar porosity can have different transient uptake because their inlet-connected void paths differ. A network-accessibility descriptor should therefore predict effective transport better than porosity alone.

This is a hypothesis, not a known result. The objectives are to:

1. quantify packing-to-packing variation under otherwise identical conditions;
2. build a conservative pore-network reference model for CO₂ diffusion and adsorption;
3. reduce every reference curve to one effective diffusivity in a 1D model; and
4. derive and test an interpretable closure from pore accessibility to effective diffusivity.

3 Proposed hybrid modelling framework

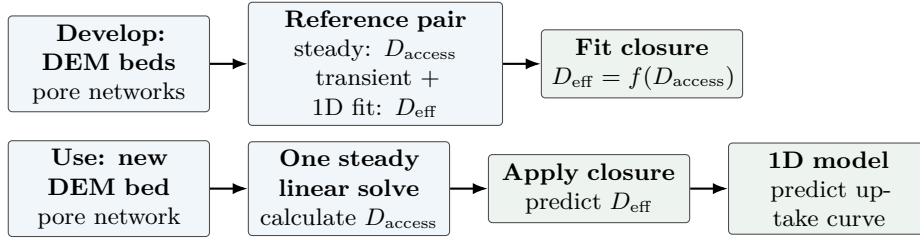


Figure 2: Two separate stages. During development, detailed simulations provide paired values of D_{access} and D_{eff} for fitting the closure. For a new bed, the fitted closure supplies D_{eff} to the fast 1D adsorption model; no transient pore-network simulation is required.

WP1: Controlled packing ensemble. Generate independent cylindrical beds of identical zeolite beads by DEM. Keep the particle number, bead size, tube size and operating conditions fixed; change only the random insertion seed. Check settling, overlaps, bed height and local and bulk porosity.

WP2: Structure-resolved reference model. Convert each void space into connected pore bodies and throats. Solve molecular diffusion on this network and couple gas loss exactly to Langmuir equilibrium and linear-driving-force uptake of the particles. Verify connectivity, void volume, numerical convergence and total mass conservation.

WP3: Define the target and the structural predictor. For each bed, obtain the reference D_{eff} by first running the full transient pore-network diffusion-adsorption model and then repeatedly solving the 1D model while fitting its transport coefficient. Thus, D_{eff} is the *target bed-scale coefficient*. Separately, impose a unit concentration difference between the inlet and an interior plane. One steady linear network solve gives the conductance G_{access} [m³ s⁻¹] and the structural descriptor

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

Here, H is the bed height and A_{tube} is the tube cross-sectional area. Unlike D_{eff} , D_{access} needs no time integration, adsorption calculation or parameter optimization. It is therefore much faster and measures how easily the inlet is connected to the interior.

WP4: Closure and predictive test. Develop and compare three mappings for the same target, D_{eff} :

$$\text{porosity: } D_{\text{eff}} = a_0 + a_1 \varepsilon_b, \quad \text{proportional: } D_{\text{eff}} = \alpha D_{\text{access}}, \quad \text{power law: } \frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Here, ε_b is bed porosity, D_m is molecular diffusivity, and the coefficients are learned from the training beds. Leave complete beds out during fitting, predict their D_{eff} , and use that value in the 1D model to predict their uptake curves. Repeat the accessibility calculation at several interior depths to check that the comparison does not depend on one selected plane.

4 Expected outcome, scope and success criterion

For a new packing, one steady accessibility solve will provide D_{access} ; the selected closure will predict D_{eff} ; and the 1D model will return the transient mean CO₂ loading. This replaces the transient pore-network calculation while retaining a link to structure. The first study isolates isothermal diffusion in stagnant gas; pressure drop, forced-flow breakthrough, heat effects and mixtures remain outside its scope. The concept succeeds if the reference model is conservative, the 1D reduction is accurate, and accessibility predicts withheld beds better than porosity. If not, the proposed structure-based closure is not justified. Either outcome provides a clear modelling decision.